

Setup Guide

Prerequisites

Install on D: drive:

- Go 1.23+
- Python 3.14+
- Node.js 20+
- PostgreSQL 18+
- Redis 7+

Database Setup

```
-- PostgreSQL
CREATE DATABASE creative_hub;

-- Create tables (run in psql)
CREATE TABLE projects (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name VARCHAR(255) NOT NULL,
  created_at TIMESTAMP DEFAULT NOW()
);

CREATE TABLE brand_kits (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  project_id UUID REFERENCES projects(id),
  colors TEXT[],
  fonts TEXT[],
  logo_url TEXT,
  created_at TIMESTAMP DEFAULT NOW()
);

CREATE TABLE assets (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  project_id UUID REFERENCES projects(id),
  type VARCHAR(50),
  content TEXT,
  url TEXT,
  metadata JSONB,
  created_at TIMESTAMP DEFAULT NOW()
);
```

Go Backend

```
cd backend-go
cp .env.example .env
# Edit .env with your credentials

go mod download
go run cmd/server/main.go
```

Access: <http://localhost:8080/health>

Python Workers

```
cd ai-workers
pip install -r requirements.txt
cp .env.example .env
# Add GROQ_API_KEY to .env

python worker.py
```

Next.js Frontend

```
cd frontend
npm install
cp .env.example .env.local

npm run dev
```

Access: <http://localhost:3000>

Testing

Text Generation:

```
curl -X POST http://localhost:8080/api/generate/text \
-H "Content-Type: application/json" \
-d '{
  "prompt": "Write a tweet about AI",
  "type": "social",
  "tone": "professional",
  "variants": 3
}'
```

Check Job Status:

```
curl http://localhost:8080/api/jobs/{job_id}
```

Troubleshooting

Redis not connecting:

- Start Redis: **redis-server**
- Check port: **redis-cli ping**

PostgreSQL issues:

- Check connection: `psql -U postgres`
- Verify database: `\l`

Go compilation errors:

- Run: `go mod tidy`
- Update deps: `go get -u ./...`

Python worker not starting:

- Check Redis: Connection should show in logs
- Verify API key in .env

Frontend build fails:

- Clear cache: `rm -rf .next node_modules`
- Reinstall: `npm install`